NIHVA
NIHVA Technologies Pvt. Ltd.

Technical Specification Sheet

Sliding Roof Material Details

1. Frame profiles and Guide Rails

The retractable roof is constructed with the combination of the 6060, 6061, 6063 grades Non corrosive Aluminium hollow extruded profiles of sectional dimension 60x60, 60x108, 50x25. Aluminium extruded guide rails 16x100. The dimensional tolerance is ±0.20.

RESULT OF ANALYSIS

Sr No	Elements	Burn 1	Burn 2	Specifications		Observed (Mean)
				Min	Max	
1	Si (%)	0.4082	0.3898	0.30	0.70	0.3990
2	Fe (%)	0.2795	0.2758	-	0.35	0.2777
3	Cu (%)	0.0303	0.0287	-	0.10	0.0295
4	Mg (%)	0.5213	0.5135	0.45	0.90	0.5174
5	Mn (%)	0.0774	0.0766	-	0.10	0.0770
6	Ti (%)	0.0098	0.0102	-	0.10	0.0100
7	Zn (%)	0.0153	0.0177	-	0.10	0.0165
8	Ni (%)	0.0000	0.0000	-	-	0.0000
9	Pb (%)	0.0000	0.0000	-	-	0.0000
10	Sn (%)	0.0000	0.0000	-	-	0.0000
11	Cr (%)	0.0095	0.0093	-	0.10	0.0094
12	Be (%)	0.0000	0.0000	-	-	0.0000
13	Sr (%)	0.0000	0.0000	-	-	0.0000
14	Zr (%)	0.0000	0.0000	-	-	0.0000
15	Ca (%)	0.0004	0.0003	-	-	0.0004
16	V (%)	0.0142	0.0148	-	-	0.0145
17	Al (%)	98.6341	98.6633	-	-	98.6487

2. Polycarbonate Sheet

Polycarbonate is one of the most advanced polymers in the field of plastics today. It offers an unequalled combination of properties: strength, transparency, light weight, flexibility, durability, thermal and fire resistance.

It is transparent as glass, 200 times stronger and less than half the weight. In addition to all of these features, its absolute resistance to breakage qualifies it as the best existing safety glazing material available, with impact resistance that is impervious to hammer blows, stones etc. It is ideal for use in areas exposed to vandalism and in cases of high impact. The sheets we use are manufactured to comply with European and U.S. standard specifications.

Both UV-A and UV-B portions of the UV spectrum are blocked out by Polycarbonate sheet. Exposure to solar ultraviolet (UV) radiation is widely known as a major health concern by now. The adverse effects were once thought to be associated with solar UV radiation in the 270-320nm (UV-B) range. However, in recent years it has become apparent that exposure to UV-A (320-400nm) is also detrimental. In addition to skin cancer, premature aging has been associated with exposure to UV-A.

- **Service Temperature**

The temperature range over which the characteristics are retained extends from -50°C to +120°C (-60°F to +250°F) for short periods and from -50°C to +100°C (-60°F to +210°F) for long periods. This range of temperatures makes polycarbonate sheets suitable for use in most climates.

Property	Method**	Conditions (U.S. Customary)*	Units - SI (U.S. Customary)*	Value (U.S. Customary)*
Physical				
Density	D-792		g/cm3 (lb/ft3)	1.2 (75)
Water Absorption	D-570	24 hr. @ 23°C	%	0.15
Mechanical				
Tensile strength at yield	D-638	10 mm/min (0.4 in./min)	MPa (psi)	62.5 (9,100)
Tensile strength at break	D-638	10 mm/min (0.4 in./min)	MPa (psi)	65 (9,500)
Elongation at yield	D-638	10 mm/min (0.4 in./min)	%	6
Elongation at break	D-638	10 mm/min (0.4 in./min)	%	>80
Tensile Modulus of Elasticity	D-638	1 mm/min (0.4 in./min)	MPa (psi)	2,300 (290,000)
Flexural Modulus	D-790	1.3 mm/min (0.052 in./min)	MPa (psi)	2,350 (343,000)
Flexural Strength at Yield	D-790	1.3 mm/min (0.052 in./min)	MPa (psi)	93 (13,600)
Notched Impact Strength Izod	D-256	23°C (73°F)	J/m (ft-lbf/in.)	800 (15)
Notched Impact Strength Charpy	D-256	23°C (73°F)	J/m (ft-lbf/in.)	800 (15)
Impact Falling Weight	ISO-6603/1b		J (ft-lbf)	158 (117)
Rockwell Hardness	D-785		R scale / M scale	125 / 75
Thermal				
Long Term Service Temperature			°C (°F)	-50 to +100 (-175 to +212)
Short Term Service Temperature			°C (°F)	-50 to +120 (-175 to +250)
Heat Deflection Temperature	D-648	Load: 1.82 MPa (264 psi)	°C (°F)	135 (275)
Vicat Softening Temperature	D-1525	Load: 1 kg (2.2 lb)	°C (°F)	150 (300)
Coefficient of Linear Thermal Expansion	D-696		mm/m °C (Mil/in. °F)	0.065 (0.036)
Thermal Conductivity	C-177		W/m K (Btu·in/hr·ft²·°F)	0.21 (1.46)
Specific Heat Capacity	C-351		kJ/kg·°K (Btu/lb·°F)	1.26 (0.31)
Optical				
Haze	D-1003	Clear Sheet	%	<0.5
Light Transmission	D-1003	Clear Sheet	%	89
Refractive Index	D-542	Clear Sheet		1.586
Yellowness Index	D-1925	Clear Sheet		<1
Electrical				
Dielectric Constant	D-150	50 Hz		3.0
	D-150	1 MHz		2.9
Dissipation Factor	D-150	1 KHz		0.001
	D-150	1 MHz		0.01
Dielectric Strength Short Time	D-149	500 V/s	kV/mm (V/mil)	>30 (>770)
Surface Resistivity	D-257	Keithley	Ohm	10^{16}
Volume Resistance	D-257	Keithley	Ohm-cm	10^{17}

3.Stainless steel fasteners

High grade Stainless steel SS304 fasteners are used to work against the harsh environmental conditions. Its high quality and durability ensure that your product lasts a long time and the design of it is suitable for home and commercial use.

4. Fabricated Parts

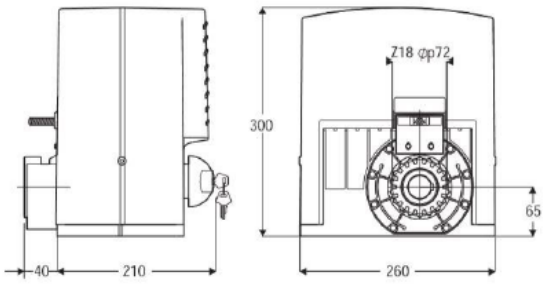
The fabricated parts are of AL standard grade accompanied with Stainless steel parts, Nylon, Teflon.

5. Ethylene Propylene Diene Monomer Synthetic Rubber Beadings and Gaskets

The sealing of all the constituents are done by specially designed high-grade Ethylene Propylene Diene Monomer beadings and gaskets. Their excellent sealing properties ensure the enclosures 100% leak proof. Their excellent resistance to heat, oxidation, ozone and weather aging due to their stable, saturated polymer backbone structure. This ensures 100% water resistant enclosures.

6. Motors with remote control and push button operations

- Rack and Pinion arrangement
- Single Phase Bi directional motor
- Speed: 09 - 12 m/min
- Power supply :230V Ac,50 hz
- Power: ¼, ½ hp (Depending on requirement)
- Pinion: Z16/Module 4/Module 6
- SS304 Mounting Base Built in control board.
- Operation: Through Remote control.
- Control Panel for Synchronization
  (Motor dimension may vary as per the capacity)



7. Additional Features

- Rain Sensor
  Rain water sensor detects the water and signals the automation to close the roof so that to avoid water entering inside the premises.

- Limit Switch
  Limit switch is a switch operated by the motion of a machine part or presence of an object. They are used for controlling machinery as part of a control system, as a safety interlock.

- Mobile App Based Functioning
  To make it more hassle free, we are introducing Mobile App based functioning which helps you open/close the roof while sitting in any part of the world.